PREDICTIVE MODELING FOR COURSE DEMAND AND REVENUE FORECASTING ON EDUPRO

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Final report on

“Predictive Modeling for Course Demand and Revenue Forecasting on EduPro”

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ABSTRACT

The EduPro Analytics dashboard is an interactive Streamlit-based web application developed to analyze online course performance, predict course demand, and forecast revenue for an educational platform. The project integrates data analytics, machine learning, and visualization techniques to provide actionable insights for decision-making. The dashboard processes and combines data from multiple datasets including courses, teachers, users, and transactions to generate meaningful business intelligence.

The system performs data preprocessing, feature engineering, and predictive modelling using Python libraries such as Pandas, NumPy, Scikit-learn, Plotly, and Streamlit. Important features such as course price, duration, level, instructor experience, and ratings are utilized to predict course revenue and enrolment demand. Machine learning models including Random Forest Regressor are implemented to evaluate and forecast course performance with accuracy metrics such as R^2 Score and MAE.

The dashboard provides several interactive modules including KPI monitoring, revenue forecast visualizations, feature importance analysis, category-level demand comparison, and a course demand prediction simulator. Users can apply filters and customize inputs to explore insights dynamically. The project demonstrates practical applications of data science and machine learning in the e-learning industry while delivering a professional and user-friendly analytical dashboard for business intelligence and strategic planning.

INTRODUCTION

1.1 About The Company

Unified Mentor, a global educational technology platform was established in 2022 with a mission to elevate employability through tailor-made skill development programs. Their offerings encompass a wide array of online courses, certification programs and internships designed to empower individuals in their quest and progress in the realms of technology, data, and marketing.

1.2 About The Project

The EduPro Analytics Dashboard is a machine learning and data analytics project developed to analyze online course performance and support data-driven decision-making for educational platforms. The project was built using Python, Streamlit, Pandas, NumPy, Plotly, and Scikit-learn to create an interactive and professional web-based dashboard. The primary objective of the project is to predict course demand, forecast revenue, and identify the key factors influencing course success.

The application integrates multiple datasets such as courses, teachers, users, and transaction records to generate meaningful business insights. Extensive data preprocessing and feature engineering techniques were applied to handle missing values, create derived attributes, and improve model performance. Features such as course price, duration, course level, instructor experience, teacher rating, and course category were used for predictive analysis.

The dashboard includes interactive modules for revenue forecasting, category-level demand comparison, KPI monitoring, and feature importance exploration. Users can apply filters and customize inputs dynamically to analyze course performance in real time. A machine learning model based on Random Forest Regressor was implemented to predict revenue and demand with strong evaluation metrics. Overall, the project demonstrates the practical implementation of data science, machine learning, and business intelligence concepts in the e-learning domain through an interactive Streamlit dashboard.

1.3 Objectives And Deliverables

The EduPro Analytics Dashboard project was developed to address the challenges faced by EduPro in understanding course performance, predicting enrolment demand, and forecasting revenue. The organization lacked predictive models for estimating course enrolments, revenue forecasting mechanisms at both course and category levels, and quantitative insights to support strategic decisions related to course launches and pricing. To overcome these limitations, the project aimed to build a data-driven analytical system using machine learning and interactive visualization techniques.

The primary objective of the project was to design an intelligent dashboard capable of analyzing educational platform data and generating meaningful business insights. The system utilizes factors such as course price, duration, course level, instructor experience, and ratings to predict demand and revenue trends. Another objective was to identify the most influential features affecting course performance through feature importance analysis.

The project deliverables include an interactive Streamlit dashboard, revenue forecasting visualizations, category-level demand comparison charts, KPI monitoring modules, and a machine learning-based prediction system. The dashboard also provides dynamic filtering capabilities and a course demand prediction simulator for real-time analysis. Overall, the project delivers a professional business intelligence solution that supports data-driven decision-making, improves strategic planning, and enhances operational efficiency for the EduPro online learning platform.

METHODOLOGY

2.1 Flow of the project

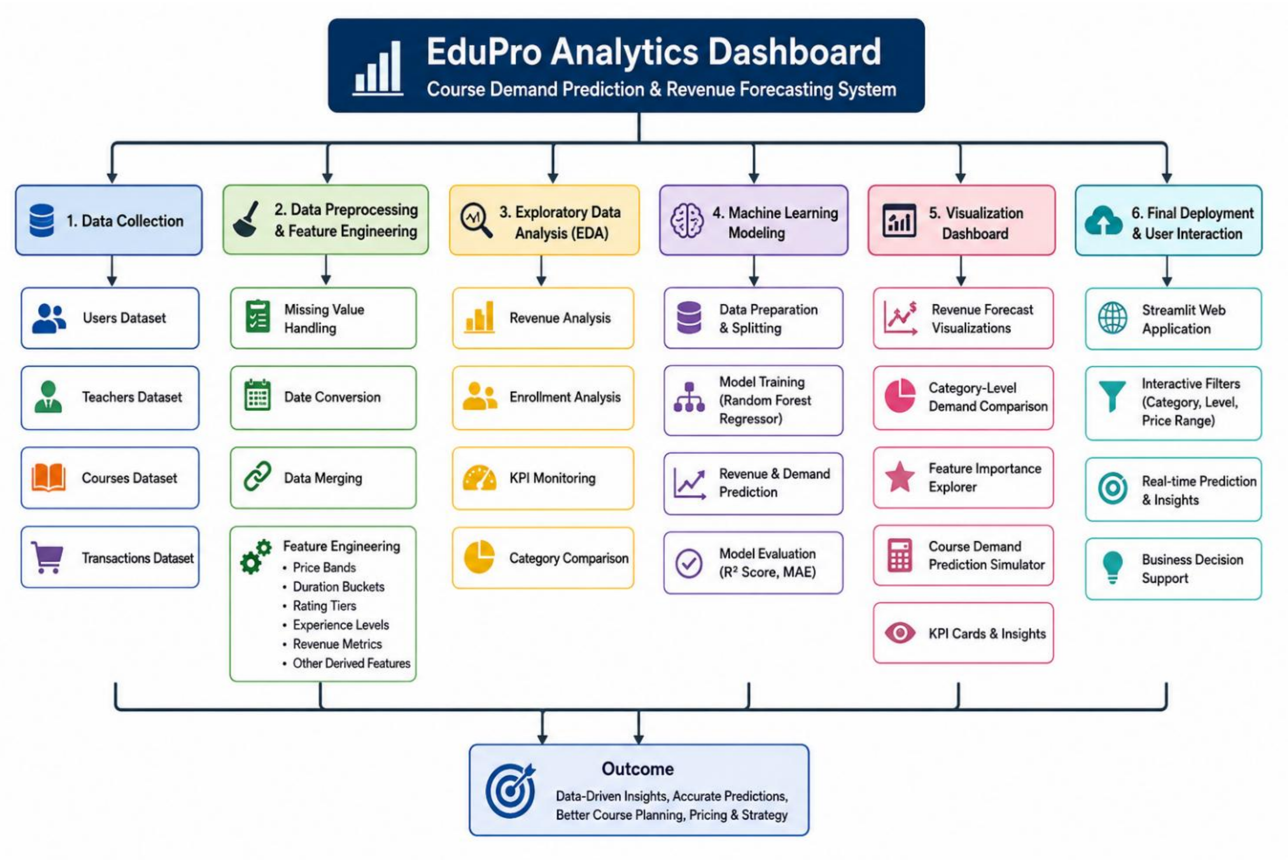
The EduPro Analytics Dashboard project follows a structured data analytics and machine learning workflow to transform raw educational platform data into meaningful business insights and predictive outputs. The project begins with data collection from multiple Excel sheets containing information related to users, teachers, courses, and transactions. These datasets are imported and integrated using Python libraries such as Pandas and NumPy.

In the next stage, data preprocessing and cleaning are performed to handle missing values, convert date columns, and prepare the dataset for analysis. Feature engineering techniques are then applied to create additional variables such as price bands, duration categories, rating tiers, experience levels, and revenue-related metrics. These engineered features improve the performance and accuracy of the predictive models.

After preprocessing, exploratory analysis and visualization modules are developed to analyze revenue trends, enrolment demand, and category-level performance. Machine learning models such as Random Forest Regressor are trained using Scikit-learn to predict course revenue and demand based on factors like course price, duration, ratings, and instructor experience. Model evaluation metrics including R2 Score and MAE are used to assess prediction accuracy.

Finally, the complete system is deployed as an interactive Streamlit dashboard with filters, KPI cards, prediction simulators, and visualization charts, enabling users to perform real-time analysis and make data-driven business decisions efficiently.

Flow Of the Project Diagram:



2.2 Use Case Diagram

The EduPro Analytics Dashboard is designed to support educational platform management through an interactive and intelligent analytical system. The primary users of the system include business analysts, administrators, and decision-makers who require insights into course performance, revenue trends, and enrolment demand. The use of case diagram of the project represents the interaction between users and the dashboard functionalities.

The system allows users to access and analyze multiple modules through a centralized Streamlit web application. Users can view key performance indicators such as total revenue, enrolments, average ratings, and course performance metrics. The dashboard enables users to apply filters based on course category, level, and price range for customized analysis. Another important use case is revenue forecasting and course demand prediction, where machine learning models generate predictive insights using factors such as course price, duration, instructor experience, and ratings.

The application also provides feature importance analysis ti help users understand which variables most strongly influence course success. Users can explore category-level demand comparisons and interactive visualizations to support strategic planning and pricing decisions. Overall, the use case diagram highlights how the dashboard transforms raw educational data into actionable insights, enabling efficient business intelligence, real-time monitoring, and data-driven decision-making for the EduPro online learning platform.

2.3 Language and Platform Used

The EduPro Analytics Dashboard project was developed using Python as the primary programming language due to its strong capabilities in data analysis, machine learning, and web application development. Python libraries such as Pandas and NumPy were used for data manipulation, preprocessing, and feature engineering. These libraries helped in handling missing values, transforming datasets, creating derived features, and performing numerical computations efficiently.

```python
import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error

```

For machine learning and predictive analytics, the Scikit-learn library was utilized to build and evaluate models for course demand prediction and revenue forecasting. Algorithms such as Random Forest Regressor were implemented to analyze patterns in course performance and generate accurate predictions. Evaluation metrics including R^2 Score and Mean Absolute Error (MAE) were used to assess model accuracy and performance.

The frontend and interactive web application were developed using Streamlit, which enabled the creation of a professional dashboard with real-time user interaction, dynamic filters, KPI cards, and prediction simulators. Plotly was used for generating interactive and visually appealing charts and graphs such as revenue forecasts, demand comparisons, and feature importance visualizations.

The project was developed and executed using platforms and tools including Jupyter Notebook, Visual Studio Code, and Streamlit Clout/ Localhost environment for testing and deployment. Overall, the combination of Python, Streamlit, and machine learning libraries provided an efficient platform for building a scalable and interactive business intelligence dashboard.

IMPLEMENTATION

3.1 Gathering Requirements and Defining Problem Statement

The EduPro Analytics Dashboard project was initiated to address the growing need for data driven decision-making in online education platforms. During then requirement growing need for data-driven objective was to understand the operational challenges faced by EduPro in managing course performance, revenue tracking, and enrolment analysis. Discussions focused on identifying the limitations of the existing system and determining how analytics and machine learning could improve strategic planning and business performance.

The analysis revealed that EduPro lacked a centralized analytical system capable of monitoring course demand, predicting revenue, and evaluating the impact of important factors such as pricing, course duration, instructor experience, and ratings. The platform also required a professional and interactive dashboard that could present insights visually and support real-time analysis for administrators and business analysts.

The project requirements included integrating multiple datasets such as users, teachers, courses, and transactions into a single analytical framework. Another major requirement was the implementation of predictive models that could forecast enrolment demand and course revenue using historical data. Additionally, the dashboard needed interactive filtering capabilities, KPI monitoring, category-level comparisons, and feature importance analysis to support decision-making.

EduPro currently lacks:

- **Predictive models for course enrolment demand**
- **Revenue forecasting at course and category level**
- **Quantitative evidence to support course launch and pricing decisions**

To solve these challenges, the project aimed to develop an interactive Streamlit-based analytics dashboard integrated with machine learning models. The final system provides revenue forecasting, demand prediction, visualization modules, and business intelligence insights, enabling EduPro to make accurate, data-driven decisions related to course planning, pricing strategies, and performance optimization.

3.2 Data Collection and Importing

The data collection and importing phase of the EduPro Analytics Dashboard project played a crucial role in building the foundation for analysis, visualization, and predictive modelling. The project required integrating multiple datasets related to the operations of the EduPro online learning platform. The collected data provided information about users, teachers, courses, and transaction records, enabling comprehensive analysis of course performance and revenue generation.

The datasets were stored in an Excel workbook containing multiple sheets, including Users, Teachers, Courses, and Transactions. Python was used as the primary programming language for importing and processing the data. The Pandas library was utilized extensively to read the Excel sheets into separate DataFrames using the read_excel() function. This approach enabled efficient handling of structured data and simplified further preprocessing operations.

```
# =========================================================
# LOAD DATASET
# =========================================================


xlsx_path = r"C:\Users\HP\Downloads\EduPro Online Platform.xlsx"

users_df = pd.read_excel(xlsx_path, sheet_name='Users')
teachers_df = pd.read_excel(xlsx_path, sheet_name='Teachers')
courses_df = pd.read_excel(xlsx_path, sheet_name='Courses')
transactions_df = pd.read_excel(xlsx_path, sheet_name='Transactions')
```

The imported datasets included important attributes such as course categories, course prices, course duration, instructor ratings, teacher experience, transaction amounts, and enrolment details. These datasets were later merged to establish relationships between courses, teachers, and transaction performance. The transaction dataset was especially important because it contained historical purchase and enrolment records that were used for revenue forecasting and demand prediction.

During the important stage, date columns such as transaction dates were converted into proper datetime formats using the to_datetime() function to support time-based analysis. Initial exploratory checks such as viewing dataset heads, identifying missing values, and understanding column structures were also performed to ensure data quality and consistency.

```

transactions_df['TransactionDate'] = pd.to_datetime(
    transactions_df['TransactionDate'],
    errors='coerce'
)

```

Overall, the data collection and importing phase established a centralized and well-structured analytical dataset that supported the development of machine learning models, interactive visualizations, KPI tracking, and predictive analytics within the Streamlit dashboard. This phase ensured that the project had accurate and organized data for generating meaningful business insights and strategic recommendations.

3.3 Designing Databases

The database designing phase of the EduPro Analytics Dashboard project focused on organizing and integrating educational platform data into a structured analytical framework. The project utilized multiple datasets containing information related to users, teachers, courses, and transactions. These datasets were designed to maintain relationships between different entities and support efficient analysis, reporting, and predictive modelling.

The courses dataset stored important information such as course category, course level, duration pricing, and course ratings. The teachers dataset contained instructor-related attributes including expertise, years of experience, and transaction dates, which are essential for revenue analysis and demand forecasting. The Users dataset-maintained learner-related information that contributed to platform activity tracking.

A relational structure was established using common identifiers such as CourseID and TeacherID to merge datasets effectively. This enabled the creation of a unified modelling dataset that combined course performance, teacher details, and transaction history. Additional derived attributes such as price bands, duration buckets, rating tiers, and experience categories were also designed during the database preparation process to improve analytical insights and machine learning performance.

The structured database design supported efficient data preprocessing, visualization, KPI generation, and predictive analytics within the Streamlit dashboard. Overall, the database framework ensured data consistency, scalability, and seamless integration for building an interactive business intelligence and course demand prediction system.

3.4 Data Cleaning

The data cleaning phase of the EduPro analytics Dashboard project was one of the most important stages in preparing the datasets for accurate analysis, visualization, and machine learning modelling. Since the project involved multiple datasets related to users, teachers, courses, and transactions, proper cleaning and preprocessing were necessary to ensure data consistency, reliability, and quality. Python libraries such as Pandas and NumPy were extensively used during this process to handle missing values, transform data, and create a structured analytical dataset.

The first step in the cleaning process involved importing the datasets from multiple Excel sheets into Pandas DataFrames using the read_excel() function. After loading the data, initial inspections were performed using functions such as head(), describe(), and column analysis to understand the dataset structure and identify inconsistencies or null values. The transaction date column was then converted into a proper datetime format using the pd.to_datetime() function with errors = ‘coerce’ to handle invalid date entries safely.

```
xlsx_path = r'"C:\Users\HP\Downloads\EduPro Online Platform.xlsx'

users_df = pd.read_excel(xlsx_path, sheet_name='Users')
teachers_df = pd.read_excel(xlsx_path, sheet_name='Teachers')
courses_df = pd.read_excel(xlsx_path, sheet_name='Courses')
transactions_df = pd.read_excel(xlsx_path, sheet_name='Transactions')

print(users_df.head())
print(teachers_df.head())
print(courses_df.head())
print(transactions_df.head())
```

Several missing values were identified in important columns such as course ratings, teacher ratings, years of experience, enrolment counts, and revenue-related attributes. Numerical missing values were handled using the fillna() function. Revenue and enrolment-related columns were filled with zero values where data was unavailable, while columns such as CourseRating, TeacherRating, and YearsOfExperience were filled using median values to maintain statistical balance and reduce the impact of outliers.

```
fill_cols = [
    'enrollment_count',
    'course_revenue',
    'avg_transaction_amount',
    'active_days',
    'revenue_per_enrollment'
]

for col in fill_cols:
    model_df[col] = model_df[col].fillna(0)

model_df['CourseRating'] = model_df['CourseRating'].fillna(
    model_df['CourseRating'].median()
)

model_df['TeacherRating'] = model_df['TeacherRating'].fillna(
    model_df['TeacherRating'].median()
)

model_df['YearsOfExperience'] = model_df['YearsOfExperience'].fillna(
    model_df['YearsOfExperience'].median()
)
```

Data aggregation techniques were also applied to clean and organize transactional information. The groupby() and agg() functions were used to calculate enrolment counts, course revenue, average transaction amounts, and transaction activity durations for each course. Duplicate and inconsistent teacher-course mappings were resolved by identifying the most frequent teacher assignment using the mode() function.

```
course_perf_df = transactions_df.groupby('CourseID').agg(
    enrollment_count=('TransactionID', 'count'),
    course_revenue=('Amount', 'sum'),
    avg_transaction_amount=('Amount', 'mean'),
    first_transaction_date=('TransactionDate', 'min'),
    last_transaction_date=('TransactionDate', 'max')
).reset_index()
```

Feature engineering was another important part of the data cleaning stage. New variables such as price bands, duration buckets, rating tiers, and instructor experience categories were created using functions like pd.qcut() and pd.cut(). Additional features such as revenue per enrolment, active course days, month of first sale, and quarter of first sale were generated to improve analytical depth and machine learning performance.

```
model_df['price_band'] = pd.qcut(
    model_df['CoursePrice'].rank(method='first'),
    3,
    labels=['Low', 'Medium', 'High']
)

model_df['duration_bucket'] = pd.qcut(
    model_df['CourseDuration'].rank(method='first'),
    3,
    labels=['Short', 'Medium', 'Long']
)

model_df['rating_tier'] = pd.cut(
    model_df['CourseRating'],
    bins=[0, 2.5, 4.0, 5.0],
    labels=['Low', 'Medium', 'High'],
    include_lowest=True
)

model_df['experience_bucket'] = pd.cut(
    model_df['YearsOfExperience'],
    bins=[-1, 3, 7, 50],
    labels=['Junior', 'Mid', 'Senior']
)
```

Overall. The data cleaning process transformed raw and unstructured educational platform data into a well-organized and high-quality dataset suitable for predictable modelling, dashboard visualizations, KPI monitoring, and business intelligence analysis within the Streamlit application.

3.5 Data Filtering

The data filtering section of the EduPro Analytica Dashboard project was implemented to provide users with an interactive and customized analytical experience. Data filtering allows business analysts and administrators to explore course performance, revenue trends, and enrolment insights based on specific conditions and user preferences. This functionality was developed using Streamlit sidebar components, enabling real-time filtering and dynamic dashboard updates.

The dashboard includes filters for important attributes such as course category, course level, and course price range. The multiselect() function in Streamlit was used to allow users to select one or multiple course categories and levels for comparative analysis. A slider() component was also implemented to filter courses within a selected price range. These filters help users focus only on relevant data and simplify the analysis of large datasets.

Filtered datasets were generated using conditional statements and logical operations in Pandas. The filtering process dynamically updates KPI metrics, charts, visualizations, and prediction outputs based on the selected inputs. This approach improves dashboard usability and enables users to analyze revenue distribution, enrolment demand, and course performance more efficiently.

The filtering system also supports better business decision-making by helping users identify high-performing categories, evaluate pricing trends, and compare demand across different course levels. Overall, the data filtering module enhanced the interactivity, flexibility, and user experience of the Streamlit dashboard while supporting accurate and targeted business intelligence analysis.

```
selected_category = st.sidebar.multiselect(
    "Select Course Category",
    options=model_df['CourseCategory'].unique(),
    default=model_df['CourseCategory'].unique()
)

selected_level = st.sidebar.multiselect(
    "Select Course Level",
    options=model_df['CourseLevel'].unique(),
    default=model_df['CourseLevel'].unique()
)

price_range = st.sidebar.slider(
    "Course Price Range",
    int(model_df['CoursePrice'].min()),
    int(model_df['CoursePrice'].max()),
    (
        int(model_df['CoursePrice'].min()),
        int(model_df['CoursePrice'].max())
    )
)
```

3.6 Prototyping – Streamlit Dashboard

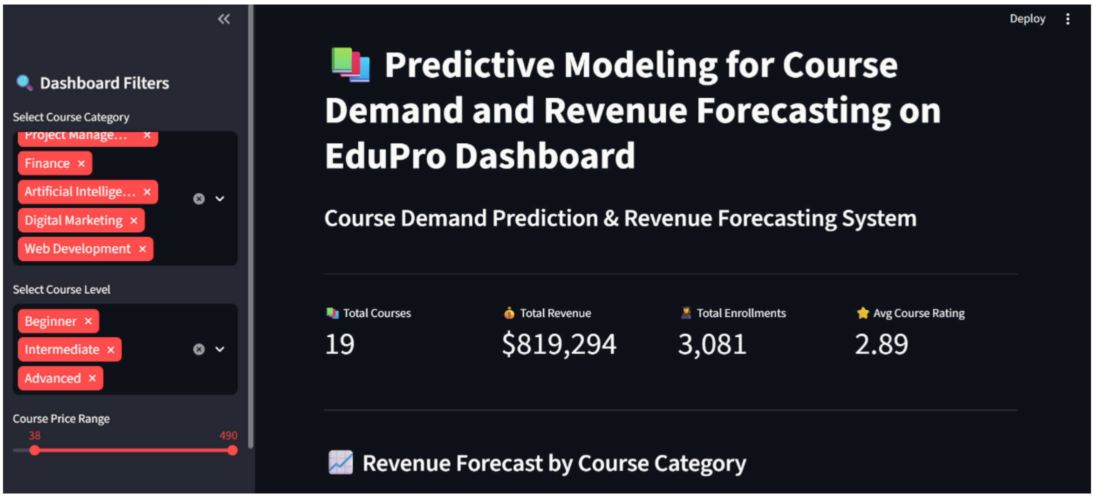
A prototype is an early version, model, or release of a product that is constructed to test a design or process. It is generally used by system analysts and users to assess a new design to enhance precision. Prototyping serves to specify a real, working system rather than a theoretical one. Creation of a prototype in some design workflow models is the step between formalizing and testing an idea.

The prototyping phase of the EduPro Analytics Dashboard project focused on designing and developing an initial working model of the interactive Streamlit application before final deployment. The main objective of this phase was to create a user-friendly interface that could effectively display course analytics, revenue forecasts, demand predictions, and business insights in a visually organized manner. The prototype helped in testing dashboard functionality, improving layout design, and validating user requirements.

The dashboard prototype was developed using Streamlit because of its simplicity, flexibility, and capability to build real-time analytical web applications using Python. Initial modules such as KPI cards, sidebar filters, and interactive visualizations were first implemented to test user interaction and dashboard responsiveness. Plotly charts and graphs were integrated to provide dynamic visual representations of revenue trends, category-level demand comparisons, and feature importance analysis.

The prototyping stage also included testing machine learning model integration within the dashboard. Prediction simulators were designed to allow users to input course price, duration, level, instructor experience, and ratings to generate revenue predictions dynamically. Different layouts, chart styles, and filtering options were evaluated to improve usability and presentation quality.

Feedback from testing and observation helped refine the overall structure, improve navigation, and optimize performance. Overall, the prototyping phase played an important role in transforming analytical requirements into a functional and professional dashboard interface, ensuring smooth interaction, better visualization, and efficient business intelligence analysis within the EduPro platform.



Exploratory Data Analysis

4.1 Defining Visuals

The defining visuals phase of the EduPro Analytics Dashboard project focused on designing clear, interactive, and visually appealing graphical components to improve data understanding and business analysis. Since the project involved large educational datasets and predictive insights, effective visualization techniques were essential for presenting complex information in a simple and meaningful format. The primary objective of the phase was to transform raw analytical data into interactive charts, KPI indicators, and graphical dashboards that support efficient decision-making.

The dashboard visuals were developed using Plotly and Streamlit, enabling dynamic and real-time interaction within the web application. Several visualization modules were designed, including revenue forecast charts, category-level demand comparisons, KPI monitoring cards, and feature importance graphs. Bar charts were used to display course category revenue distributions, while pie charts helped visualize enrolment demand across different course categories. Feature importance visualizations were created to identify the most influential factors affecting course revenue and demand prediction.

Special attention was given to dashboard layout, colour themes, spacing, and charts positioning to ensure a professional appearance and better user experience. KPI cards displaying total revenue, enrolments, average ratings, and total courses were added to provide quick business insights at a glance. Interactive filters and prediction simulators further enhanced dashboard usability by allowing users to customize analysis dynamically.

The defining visual stage improved the presentation quality, interpretability, and interactivity of the Streamlit dashboard, making the analytical outputs more understandable, engaging, and useful for strategic business intelligence and performance monitoring.

4.2 Visualizations

The visualization section of the project presents course performance, revenue trends, enrolment demand, and feature importance through interactive charts and KPI cards. Using Plotly and Streamlit, the dashboard provides dynamic bar charts, pie charts, and forecasting visuals that help users analyze data affectively and support data-driven business decision-making.

Key Performance Indicator (KPI)

The KPI (Key Performance Indicator) section of the EduPro Analytics Dashboard provides a quick summarized overview of the overall performance of the online learning program. This section was designed to help administrators and business analysts monitor important metrics in real time and make informed business decisions efficiently. Using Streamlit metric cards, the dashboard presents key insights in a visually clear and professional format.

The KPI module displays major indicators such as Total Courses, Total Revenue, Total Enrolmets, and Average Course Rating. These metrics are dynamically updated based on user-selected filters such as course category, level, and price range. The KPI cards improve dashboard usability by presenting high-level business information at a glance without requiring detailed analysis of raw datasets.

The Total Courses metric helps measure the number of active courses available on the platform, while Total Revenue indicates the overall income generated from course enrolments and transactions. Total Enrolments reflects kearner participation and demand across different course categories. The Average Course Rating metric evaluates overall course quality and student satisfaction levels. The KPI section was developed using Streamlit components and integrated with Pandas-based filtered datasets for real-time updates.

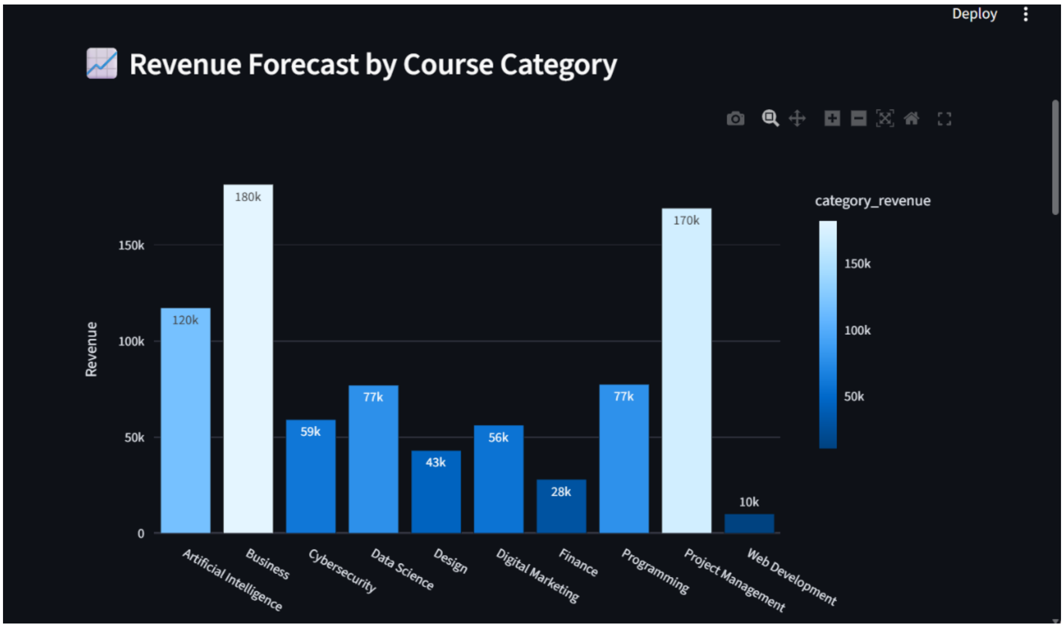
The KPI section enhances the business intelligence capabilities of the dashboard by providing concise performance summaries, supporting trend analysis, and enabling quick strategic valuation of the EduPro platform.

Observation from the KPI Dashboard Image:

- The platform curently offers **19 courses** across different vategories and levels.
- The dashboard shows a **high total revenue of $819,294**, indicating strong financial performance and successful course monetization.

- A total number of **3,081 enrolments** suggests good learner engagement and consistent demand for courses.
- The **average course rating is 2.89**, which is reletively moderate and indicates that there is scope for improving course quality, learner satisfaction, and teaching effectiveness.
- The comparison between high revenue and moderate ratings suggests that while courses are generating strong income, improving course content and user experience could further increase retemtion and platform reputation.

Revenue Forecast By Course Category



The “Revenue Forecast by Course Category” chart represents the total predicted or generated revenue for different course categories available on the EduPro platform. The var chart helps compare the financial performance of each category and identify which domains contribute the most to the platform’s overall revenue. The x-axis represents different course categories, while the y-axis represents the revenue generated in each category. The color intensity of the bars also indicates the magnitude of revenue, where lighter bars represent higher revenue values.

This visualization was created using Plotly within the Streamlit dashboard to provide an interactive and business-friendly analysis interface. The chart enables administrators and analysis to identify profitable course categories, evaluate market demand, and make strategic related to pricing promotions, and future course launches.

Obseravtion from the chart:

Business Category Generates The Highest Revenue

- **The Business Category** shows the highest revenue at approximately **180k**.
- This indicates strong learner demand and high profitability in business related courses.
- The platform may focus on expanding business courses to maximize revenue growth.

Project Management Is Thesecond Highest Performer

- Project management generates around **170k** revenue.
- This suggests that professional and career-oriented courses are highly popular among users.

Artificial Intelligence performs strongly

- The **Artificial Intelligence** category contributes approximately **120k** revenue.

- The high revenue reflects gorwing market demand for AI and emerging technology skills.

Mid-Level Revenue Categories

- **Data Science** and **Programming** each generate nearly **77k** revenue.
- **Cybersecurity** and **Digital Marketing** contribute around **56k-59k**.
- These categories show stable performance and moderate learner engagement.

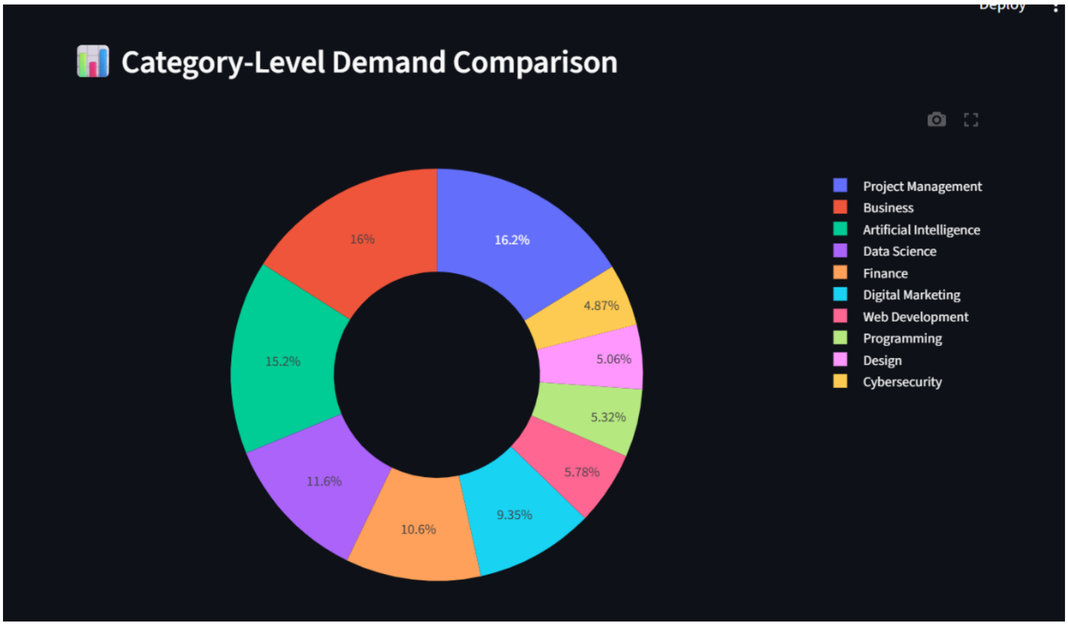
Low Revenue Categories

- **Finance** generates only about **28k**, while **Web Development** contributes the lowest revenue at approximately **10k**.
- These categories may require better marketing strategies, updated content, or pricing improvements to increase enrolments.

Overall Business Insight

- Technical and professional development courses dominate the revenue contribution.
- High-demand categories such as Business AI, and Project Managemnet should be priortized for future investment and expansion.
- Lower-performing categories can be improved through targeted promortions, improved course quality, or revised pricing strategies.

Category-Level Demand Comparison



The "Category-Level Demand Comparison" chart is a donut pie chart that represents the percentage distribution of student enrolment demand across different course categories on the EduPro platform. Each segment of the chart indicates the share of the total enrolments contributed by a particular course category. The chart helps visualize learner internet, market trends, and the popularity of various domains offered on the platform.

The visualization was developed using Plotly within the Streamlit dashbaord to provide an interactive and easy-to-understand representation of enrolment demand. The donut structure improves readability by clearly separating each category while maintaining a professional dashboard appearance. Different colors are assigned to individual categories to distinguish their contribution levels effectively.

This chart enables business analysts and administrators to identify high-demand categories, understand learner preferences, and make strategic decisions regarding course expansion, marketing, pricing, and content development.

Observation From The Chart:

Project Management Has The Highest Demand

- **Project Management** contributes approximately **16.2%** of total enrolments.
- This indicates that learners highly prefer professional and career-oriented management courses.
- The platform can priortize launching more advanced project managemnet programs.

Business Courses Show Strong Learner Interest

- The **Business** category contributes around **16%** demand.

- This suggests that business-related skills remain highly valuable and widely prefered among learners.

Artificial Intelligence Is Rapidly Growing

- **Artificial Intelligence** contributes approximately **15.2%** demand.
- The high percentage reflects the increasing industry demand for AI and emerging technology skills.

Data Science and Finance Maintain Moderate Demand

- **Data Science** contributes around **11.6%,** while **Finance** contributes approximately **10.6%.**
- These categories show stable learner engagement and consistent interest.

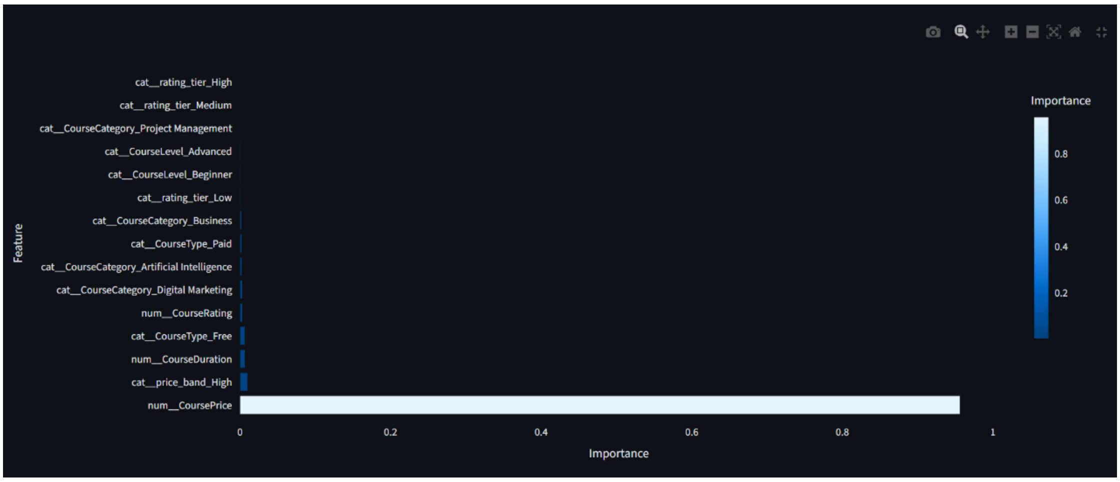
Lower Demand Categories

- **Digital Marketing** contributes around **9.35%** demand.
- **Web Development**, **Programming, Design**, and **Cybersecurity** contribute between **4% - 6%** each.
- These lower percentages may indicate market competition, limited course variety, or reduced learner engagement.

Overall Business Insight

- Professional and technology-oriented courses dominate overall enrolment demand.
- Management, Business, and AI categories together contribute a major portion of total learner interest.
- Lower-performing categories may require improved marketing strategies, updated course content, better pricing, or additional specialization courses to increase enrolments and platform engagement.

Feature Importance Explorer



The “Feature Importance Explorer” chart represents the contribution of different input features used in the machine learning model for predicting course revenue in the EduPro Analytics Dashboard. This visualization helps identify which variables have the strongest influence on revenue forecasting and course demand prediction. The chart was generated using the feature importance values obtained from the Random Forest Regressor model implemented in Scikit-learn.

In the graph, the y-axis represents the different features used in the prediction model, while the x-axis represents the importance score of each feature. Higher importance values indicate that the feature has a greater impact on the model’s prediction performance. The visualization helps business analysts and administrators understand which course-related attributes most significantly affect revenue generation.

The chart includes features such as Course Price, Price Band, Course Duration, Course Rating, Course Level, Course Category, Course Type, and Rating Tiers. These variables were engineered during preprocessing and used as model inputs for predictive analysis. The feature importance explorer enhances model interpretability and supports data-driven strategic planning.

Observation Of The Graph:

Course Price Is The Most Important Feature

- **Num_CoursePrice** has the highest importance score, nearly close to **1.0**.
- This indicates that course pricing is the strongest factor influencing revenue prediction.
- Higher priced courses significantly impact total revenue generation.

Price Band Also Influences Revenue

- **Cat_price_band_High** has noticeable importance after **Course Price.**
- This suggests that premium-priced course categories contribute more to platform revenue.

Course Duration Impacts Revenue Moderately

- **Num_CourseDuration** shows moderate importance.
- Longer-duration courses may attract higher pricing and improved revenue opprtunities.

Free Vs Paid Course Type Affects Performance

- **Cat_CourseTypr_Free** and **cat_CourseType_Paid** influence the premium model.
- Paid courses contribute more directly to revenue generation compared to free courses.

Course Ratings And Levels Have Lower Influence

- Feature such as:

Course rating

Beginner/advanced levels

Rating tiers

have relatively smaller importance scores.

- This indicates that pricing-related factors dominate revenue prediction more than learner ratings or difficulty levels.

Courses Categories Have Business Impact

- Categories such as:

Project management

Business

Artificial intelligence

Digital marketing

show measurable influence on revenue performance.

- These categories contribute to overall profitability and learner demand.

Overall Business Insight

- Revenue greneration is primarily driven by pricing strategy and closure structure.
- EduPro can improve business performance by:

Optimizing course pricing

Creating premium course packages

Expanding high-performing categories

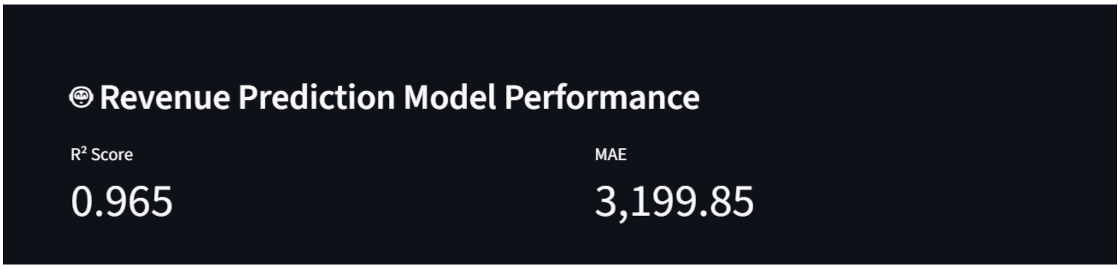
Designing longer and value-based courses

Strategic Importance Of The Visualization

- The feature importance chart improves transparancy and interpretability of the machine learning model.

- It helps stakeholders understand which business factors directly affect revenue forecasting and supports more informed strategic decisions.

Revenue Prediction Model Performance



The “Revenue Prediction Model Performance” section evaluates the accuracy and effectiveness of the machine learning model used in the EduPro Analytics Dashboard for predicting course revenue. The model was developed using the Random Forest Regressor algorithm from Scikit-learn and trained on features such as course price, duration, course level, instructor experience, teacher rating, and category information.

This performance section displays two evaluation metrics: R^2 Score and MAE (Mean Absolute Error). These metrics help determine how accurately the model predicts course revenue using historical educational platform data.

The R^2 Score measures how ell the model explains the variation in revenue data. Its value ranges from 0 to 1, where values closer to 1 indicate better prediction accuracy. The MAE metric measures the average difference between actual revenue values and predicted revenue values. Lower MAE values indicate better model performance and fewer prediction errors.

The model evaluation module was integrated into Streamlit dashboard to provide transparency, reliability, and confidence in the predictive analytics system. This helps administrators and analysts understand the effectiveness of the forecasting model before using it for business decision-making.

Observation From the Chart:

High R2 Score Indicates Excellent Model Accuracy

- The model achieved an **R^2 Score of 0.965**
- This means the model explains approximately **96.5% of the variance** in course revenue data.
- The prediction model is highly accurate and reliable for revenue forecasting.

Low Prediction Error

- The **MAE value is 3,199.85**, which represents the average prediction error.
- Compared to the total revenue values in the dataset, this error is relatively low and acceptable.

Strong Predictive Capability

- The high R^2 Score and low MAE together indicate that the Random Forest model successfully captures important relationships between course features and revenue generation.

Business Impact

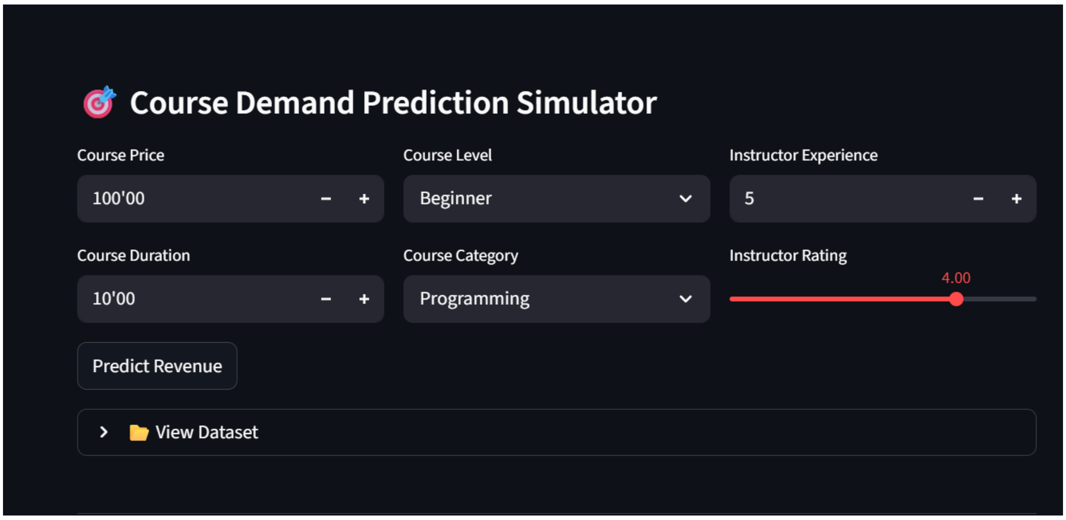
The model can effectively support **strategic business decisions** such as:

- Course Launch Planning
- Pricing Optimization
- Revenue Forecasting
- Demand Analysis
- Resource Allocation

Overall Insight

- The performance metrics demonstrate that the machine learning model is robust, efficient, and suitable for real-time predictive analytics within the EduPro dashboard.
- The system can be trusted for generating data-driven insights and improving business intelligence operations.

Course Demand Prediction Simulator



The “Course Demand Prediction Simulator” is an interactive module of the EduPro Analytics Dashboard designed to predict course revenue based on user-defined inputs. This simulator allows administrators, analysts, and decision-makers to test different course configurations and analyze how various factors influence revenue generation and course demand. The feature was developed using Streamlit input components and a machine learning prediction model integrated into the dashboard.

The simulator takes important course-related parameters such as course price, course duration, course level, course category, instructor experience, and instructor Rating as input variables. These features are processed through a trained Random Forest Regressor model, which predicts the expected revenue for the selected course configuration. The module provides real-time predictions, making the dashboard more interactive and business-oriented.

The interface includes dropdown menus, sliders, and numeric input fields that improve usability and allow users to experiment with different combinations of course attributes. A “Predict Revenue” button triggers the machine learning model and generates the predicted output instantly. The simulator also includes a dataset viewing option for transparency and further analysis.

Observation Says:

Interactive Prediction System

- The simulator allows users to dynamically modify course-related inputs and instantly receive revenue predictions.
- This improves real-time business analysis and decision-making capabilities.

Current Input Configuration

The current selected values are:

- Course Price: 100
- Course Duration: 10
- Course Level: Beginner
- Course Category: Programming
- Instructor Experience: 5 Years
- Instructor Rating: 4.0

These values represent a beginner-level programming course with moderate pricing and an experienced instructor.

Instructor Rating And Experience Influence Predictions

The instructor rating is relatively high at **4.0**, which positively impacts predicted course demand and revenue.

Instructor experience of **5** years suggests mid-level expertise, which may improve learner trust and engagement.

Business Value of The Simulator

The simulator helps EduPro test pricing strategies before launching new courses.

It supports:

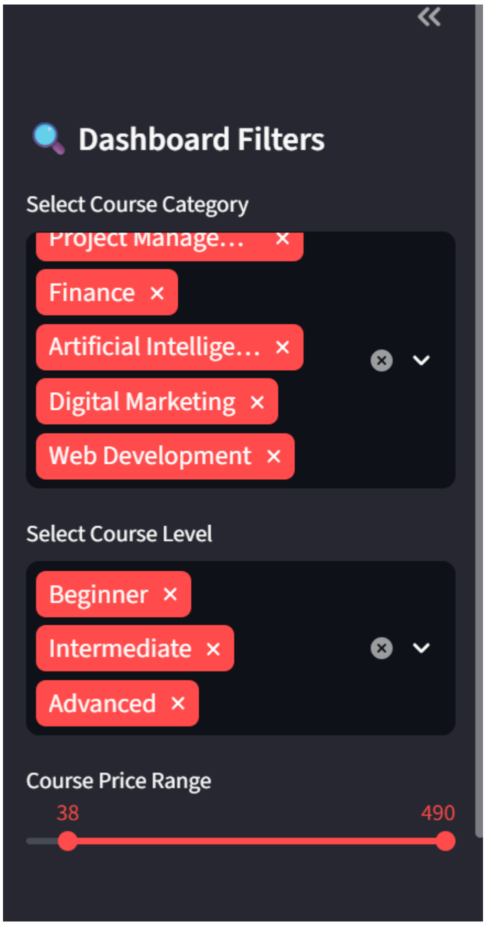
- Revenue Forecasting
- Course Planning
- Pricing Optimization
- Demand Analysis
- Strategic Decision-Making

Overall Insight

The module demonstrates the practical integration of machine learning with business intelligence.

It transforms static analytics into an interactive predictive system capable of supporting real-time educational business strategies and performance evaluation.

Dashboard Filters



The “Dashboard Filters” section is an interactive control panel in the EduPro Analytics Dashboard that allows users to customize and refine the displayed analytical results. This filtering module was developed using Streamlit sidebar components to provide dynamic and real-time data exploration. The purpose of these filters is to help users analyze specific course categories, course levels, and pricing ranges according to their business requirements.

The filter panel contains three main controls:

Select course category – allows users to choose one or multiple course categories.

Select Course Level – enables filtering courses based on difficulty levels such as Beginner, Intermediate, and Advanced.

Course Price Range – provides a sidebar to filter courses within a selected price range.

These filters directly affect the dashboard outputs, including KPI cards, revenue forecasts, demand comparison charts, and prediction modules. Whenever a filter is changed, the dashboard automatically updates all visualizations and metrics based on the selected conditions. This improves analytical flexibility and enhances user interaction with the dashboard.

Sidebar Filters Includes:

Multiple Course Categories Selected

The selected categories include:

- Project management
- Finance
- Artificial intelligence
- Digital marketing
- Web development

This indicates that the user wants to compare performance across both technical and management-oriented domains.

All Course Levels Are Included

The selected levels are:

- Beginner
- Intermediate
- Advanced

This means the dashboard analysis currently covers courses for all learner skill levels, providing a complete market overview.

Wide Price Range Selection

- The course price range is set between 38 and 490.
- This includes both low-cost and premium courses, allowing broader revenue and demand analysis.

Enhanced User Interaction

- The multi-select dropdowns and sliders make the dashboard highly interactive and user-friendly.
- Users can quickly customize insights without modifying the dataset manually.

Business Intelligence Advantage

The filtering system helps users:

- Analyze category-specific performance
- Compare learner demand across skill levels
- Evaluate pricing trends
- Identify high-performance course segments
- Make targeted strategic decisions.

Overall Insight

- The dashboard filters improve flexibility, personalization, and analytical efficiency.

- They allow EduPro administrators and analysts to perform focused business analysis and generate more accurate insights from the educational platform data.

Conclusion And Future Scope

The EduPro Analytics Dashboard project successfully demonstrates the integration of data analytics, machine learning, and interactive visualization techniques to support data-driven decision-making in the online education domain. The project was developed to address the lack of predictive systems for course demand analysis, revenue forecasting, and strategic course planning within the EduPro platform. By combining datasets related to courses, teachers, users, and transactions, the system provides meaningful insights into course performance and learner behavior.

The dashboard includes modules such as KPI monitoring, revenue forecasting, category-level demand comparison, feature importance analysis, and a course demand prediction simulator. Machine learning models, particularly the Random Forest Regressor, were implemented to generate accurate revenue predictions with strong evaluation performance. Interactive filters and visualization created using Streamlit and Plotly improved user experience and enabled real-time business analysis.

In the future, the project can be enhanced by integrating advanced deep learning models and real-time cloud-based databases for improved scalability and prediction accuracy. Additional features such as student recommendations systems, sentiment analysis from learner feedback, and personalized course suggestions can further improve platform intelligence. Deployment on cloud platforms and integration with live educational systems can also make the dashboard more efficient and accessible. Overall, the project provides a strong foundation for intelligent educational analytics and future AI-driven business solutions in the e-learning industry.

REFERENCE

The following websites have been referred for Python and Streamlit tutorials:

a. Our Documentation | Python.org
b. Streamlit documentation
c. YouTube
d. Wikipedia
e. GitHub · Change is constant. GitHub keeps you ahead. · GitHub
f. Parks & Recreation – City of Toronto